Coding & Solution

Date	1 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	5 Marks

Coding & Solution

This section presents the implementation of the **OptiCrop – AI-Based Crop Recommendation System**. The solution uses Machine Learning to recommend the most suitable crop based on soil nutrients and environmental conditions. The application is developed using **Python, Flask, HTML, CSS, and JavaScript**, following clean coding practices, modular design, and project requirements to ensure reliability, maintainability, and ease of future enhancements.

Instructions

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/farru052000-ops/OPTI_CROP_AI
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	Crop recommendation, data preprocessing, Exploratory Data Analysis (EDA), Machine Learning model training, trained ML model integration, responsive web interface, feature scaling using StandardScaler, input validation, crop prediction display.
Pending / Incomplete Features	User authentication, weather API integration, prediction history, database integration, multilingual support.
Setup / Run Instructions	Install dependencies using pip install -r requirements.txt, then run python app.py and open the application in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The **OptiCrop – AI-Based Crop Recommendation System** successfully integrates a Machine Learning model with a Flask web application to provide accurate crop recommendations based on soil nutrients and environmental conditions. The project follows a modular architecture with separate folders for **templates, static files, dataset, model, and source code**, making it easy to maintain and extend with additional features in future versions. The application has been successfully deployed on **Render**, and the complete source code is available on **GitHub**.